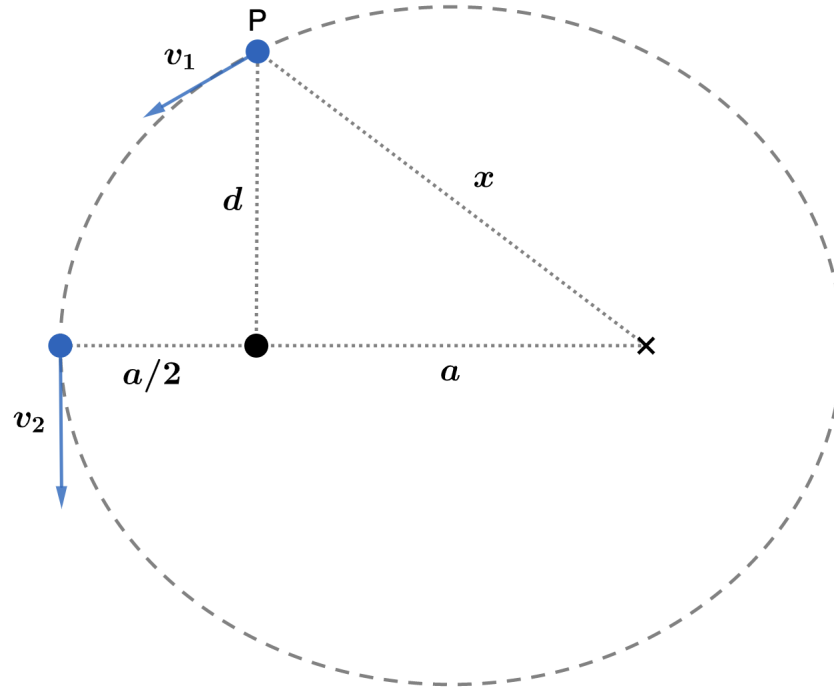


2017 F=ma Exam: Problem 25

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Recall the vis-viva equation

$$v = \sqrt{GM \left(\frac{2}{r} - \frac{1}{a} \right)}$$

gives the speed of the planet at any point on the elliptical orbit. The ratio of the speeds at two points is then

$$\frac{v_2}{v_1} = \sqrt{\frac{\frac{2}{r_2} - \frac{1}{a}}{\frac{2}{r_1} - \frac{1}{a}}}$$

At point P , we have $r_1 = d$. By the definition of an ellipse, we have

$$d + x = 2a$$

We can also use the Pythagorean theorem to get

$$d^2 + a^2 = x^2$$

Substituting this in the previous equation,

$$d^2 + a^2 = (2a - d)^2 = 4a^2 - 4ad + d^2$$

$$4ad = 3a^2$$

$$d = \frac{3a}{4}$$

At the perigee, we are given $r_2 = a/2$. Thus,

$$\frac{v_2}{v_1} = \sqrt{\frac{\frac{2}{a/2} - \frac{1}{a}}{\frac{2}{3a/4} - \frac{1}{a}}} = \sqrt{\frac{4 - 1}{(8/3) - 1}} = \sqrt{\frac{9}{5}}$$

$$v_2 = \frac{3}{\sqrt{5}}v_1$$

so the answer is $\boxed{A}$.